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October 28, 2026

The Editor-in-Chief
Measurement

Dear Editor,

We submit the manuscript *Valid uncertainty from few model runs: surrogates in GUM Supplement 1*, by Yugam Jinesh Vora (National University of Singapore) and Wee Hoe Ng (National Metrology Centre, A*STAR), for consideration as a research article.

Contribution to measurement science. When the measurement model is an expensive simulation, the Monte Carlo method of GUM Supplement 1 is applied to a surrogate fitted to a few tens of model runs, and the error of the surrogate becomes a component of the measurement uncertainty. Which surrogate to use, and how to include its error in the uncertainty budget so that the stated coverage holds, are open questions in practice. The manuscript answers both with a paired comparison on twelve models with up to 20 inputs, including GUM-type input distributions and a computational fluid dynamics simulation, at budgets of 10 to 80 runs, with every uncertainty statement checked against reference Monte Carlo evaluations of the true models. Its main findings are:

- the specification of the surrogate matters more than committees or stacking of surrogates, the type of design or adaptive designs, and from 20 runs on, choosing among a few candidates by their leave-one-out error is practically as good as the best choice in hindsight;
- adding the surrogate's resampled leave-one-out residuals to every Monte Carlo trial gives GUM S1 coverage intervals with the nominal coverage from 10 to 80 runs, also with dependent inputs and for vector outputs, whereas intervals from the Gaussian process predictive variance, the usual choice, often fell short of their nominal coverage, also with the hyperparameters treated in a fully Bayesian way;
- the results are condensed into a procedure, applied in a worked example with an uncertainty budget for a gas-dispersion simulation.

The work fits the scope of *Measurement* in the evaluation of measurement uncertainty and the mathematical tools for measurement, and is directed at metrologists and engineers who evaluate uncertainty with simulation models.

Prior presentation. A study at a budget of 20 runs on four models has been submitted as an extended abstract to the XXV IMEKO World Congress (Rimini, 2027) [CONFIRM status at submission]. The manuscript extends it substantially and has not been published or submitted elsewhere.

Data and code. All code, the per-seed result records, the cached simulation runs and a script that recomputes every number in the manuscript are openly available and archived at Zenodo (DOI on the title page). The manuscript is anonymized for double-anonymized review; the declaration on the use of generative AI is on the title page.

[CONFIRM: optional suggested reviewers, chosen by the authors, with their affiliations and e-mail addresses.]

The authors declare no competing interests. Thank you for considering the manuscript.

Yours sincerely,

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(corresponding author, on behalf of both authors)